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The ECME algorithm: A simple extension of EM and ECM with faster monotone convergence

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SUMMARY

A generalisation of the ECM algorithm (Meng & Rubin, 1993), which is itself an extension of the EM algorithm (Dempster, Laird & Rubin, 1977), can be obtained by replacing some CM-steps of ECM, which maximise the constrained expected complete-data loglikelihood function, with steps that maximise the correspondingly constrained actual likelihood function. This algorithm, which we call ECME algorithm, for Expectation/Conditional Maximisation Either, shares with both EM and ECM their stable monotone convergence and basic simplicity of implementation relative to competing faster converging methods. Moreover, ECME can have a substantially faster convergence rate than either EM or ECM, measured using either the number of iterations or actual computer time. There are two reasons for this improvement. First, in some of ECME's maximisation steps, the actual likelihood is being conditionally maximised, rather than a current approximation to it, as with EM and ECM. Secondly, ECME allows faster converging numerical methods to be used on only those constrained maximisations where they are most efficacious. Illustrative ECME algorithms are presented with both closed-form and iterative CM-steps, which demonstrate the faster rate of convergence and the associated easier assessment of convergence. Also, theoretical expressions are presented and illustrated regarding the rate of convergence of ECME. Finally, relationships with Markov chain Monte Carlo methods are discussed.

Some key words: Incomplete data; Likelihood; Maximum likelihood; Missing data.

1. INTRODUCTION

Since the presentation of the EM algorithm by Dempster, Laird & Rubin (1977), a substantial amount of work has been devoted to trying to speed EM's rate of convergence without losing its very appealing simplicity of implementation and monotone convergence properties; see, for example, Louis (1982), Laird, Lange & Stram (1987) and Lange (1995). Some of these methods, after a number of initial EM iterations intended to move to a region where the loglikelihood is nearly quadratic, shift to faster converging maximisation methods based on a quadratic approximation, such as Newton–Raphson, which require a $d \times d$ matrix of second derivatives, where d is the dimensionality of the parameter θ whose maximum likelihood estimate, or posterior mode, is sought. Even ignoring the extra analytic and algorithmic work to implement such an algorithm, the extra computational complexity can be substantial with large d , especially considering that, if used prematurely, the global quadratic approximation underlying a Newton-stepping algorithm can be inaccurate enough to lead to decreases in likelihood and erratic behaviour.

Here we present an extension of the ECM algorithm of Meng & Rubin (1993); ECM is itself an extension of EM that typically converges more slowly than EM in terms of iterations

but can be faster in total computer time. Our new extension is nearly always faster than both EM and ECM in terms of number of iterations and moreover can be faster in total computer time by orders of magnitude.

The basic idea of ECM is to replace the M-step of each EM iteration with a sequence of $S > 1$ conditional or constrained maximisation, or CM, steps, each of which maximises the expected complete-data loglikelihood found in the preceding E-step subject to constraints on θ , where the collection of all constraints is such that the maximisation is over the full parameter space of θ . That is, each CM-step maximises the expected complete-data loglikelihood over some function of θ , say $\theta_s = \theta_s(\theta)$ ($s = 1, \dots, S$), where the θ_s span the θ space. A CM-step might be in closed form or it might itself require iteration, but because the CM maximisations are over smaller dimensional spaces, often they are simpler, faster, and more reliable than the corresponding full maximisation called for in the M-step of EM, especially when iteration is required.

The same benefits of working in a lower dimensional space hold for maximising the actual likelihood function subject to the same constraints. That is, in a CM-step, we can consider maximising the actual loglikelihood function rather than the expected complete-data loglikelihood function. This suggestion leads to our ECME algorithm, where each CM-step maximises either the expected complete-data loglikelihood, as with ECM, or the actual likelihood function, subject to the same constraints on θ . The advantage of ECME over EM and ECM is a potentially dramatic increase in the speed of convergence because a CM-step that maximises the actual loglikelihood over θ_s has discarded the memory of the previous estimate of θ_s contained in the expected complete-data loglikelihood being maximised by EM or ECM. Moreover, ECME has the flexibility to use faster converging methods to maximise those conditional likelihoods for which they are most effective. Any problem that has been attacked by EM or ECM can also be attacked by ECME. Typically ECME is more tedious to code than ECM, but the reward of faster convergence is often worthwhile, especially because it allows convergence to be more easily and confidently assessed with real world finite precision computation. Furthermore, as with EM and ECM, ECME monotonely increases the likelihood and reliably converges to a maximum likelihood estimate.

Section 2 presents the basic theory on monotone convergence of ECME to a maximum likelihood estimate. Section 3 summarises numerical examples chosen to illustrate various practically important points. Section 4 summarises the underlying theory on rates of convergence, and illustrates this theory in a simple case where expressions can be easily evaluated. Finally, § 5 comments on the relationship between ECME and Markov chain Monte Carlo methods.

2. BASIC THEORY

Let $X \in \mathcal{X}$ be the complete-data with density $f(X|\theta)$, and $Y \in \mathcal{Y}$ the observed incomplete data, where $\theta \in \Theta$, and $Y = Y(X)$ is a many-to-one mapping from $\mathcal{X}$ to $\mathcal{Y}$. Also let $g(Y|\theta)$ denote the density of Y and $k(X|Y, \theta)$ the conditional density of X given Y ; then $g(Y|\theta) = \int_{\mathcal{X}(Y)} f(X|\theta) dX$, where

$$\mathcal{X}(Y) = \{X: X \in \mathcal{X}, Y(X) = Y\}, \quad f(X|\theta) = g(Y|\theta)k(X|Y, \theta).$$

The objective is to find the maximum likelihood estimate of θ , $\hat{\theta}$, which maximises the

actual loglikelihood

$$L(\theta) \equiv \log g(Y|\theta) = Q(\theta|\theta') - H(\theta|\theta'), \quad (1)$$

where

$$Q(\theta|\theta') = E\{\log f(X|\theta) | Y, \theta'\}$$

is the expected complete-data loglikelihood, and

$$H(\theta|\theta') = E\{\log k(X|Y, \theta) | Y, \theta'\}$$

is the expected missing-data loglikelihood.

The EM algorithm maximises $L(\theta)$ by iteratively maximising $Q(\theta|\theta')$ over θ . The t th iteration $\theta^{(t)} \rightarrow \theta^{(t+1)}$ of the EM algorithm is defined by an expectation, or E, step, which computes $Q(\theta|\theta^{(t)})$ as a function of θ , followed by a maximisation, or M, step, which finds $\theta = \theta^{(t+1)}$ to maximise $Q(\theta|\theta^{(t)})$. Each iteration of EM increases $L(\theta)$, and very generally, if EM converges to θ^* , θ^* is a local maximum of $L(\theta)$ (Dempster et al., 1977). Other results on the convergence of EM appear in various places, such as Wu (1983).

The ECM algorithm replaces the M-step with a sequence of simpler constrained or conditional maximisation, CM, steps, indexed by $s = 1, \dots, S$, each of which fixes some function of θ , $h_s(\theta)$. Meng & Rubin (1993) show that the ECM algorithm is a generalised EM, or GEM, algorithm (Dempster et al., 1977) and shares the nice convergence properties of EM under a 'space-filling' condition on the set of constraint functions, $\mathcal{H} = \{h_s(\theta) : s = 1, \dots, S\}$.

The ECME algorithm replaces some of ECM's CM-steps with steps that maximise the correspondingly constrained actual likelihood function $L(\theta)$. The notation required to describe precisely this simple idea is that of Meng & Rubin (1993) extended to have $s \in \mathcal{S}_Q \cup \mathcal{S}_L = \{1, \dots, S\}$.

DEFINITION. The ECME is an iterative algorithm, $\theta^{(t)} \rightarrow \theta^{(t+1)}$, consisting of an E-step, which computes $Q(\theta|\theta^{(t)})$ as a function of θ , and S constrained maximisation steps indexed by s with input $\theta^{(t+(s-1)/S)}$ and output $\theta^{(t+s/S)}$. For $s \in \mathcal{S}_Q$, $Q(\theta^{(t+s/S)}|\theta^{(t)}) \geq Q(\theta|\theta^{(t)})$ for all θ satisfying $h_s(\theta) = h_s(\theta^{(t+(s-1)/S)})$; for $s \in \mathcal{S}_L$, $L(\theta^{(t+s/S)}) \geq L(\theta)$ for all θ satisfying $h_s(\theta) = h_s(\theta^{(t+(s-1)/S)})$; $\mathcal{S}_Q \cup \mathcal{S}_L = \{1, \dots, S\}$.

Different algorithms, in the sense of different sample paths $\theta^{(0)}, \theta^{(1)}, \dots$ are obtained for different orderings of the S steps.

THEOREM 1. The ECME algorithm monotonically increases the likelihood function, $L(\theta)$.

Proof. For each $s \in \mathcal{S}_Q$, $Q(\theta|\theta^{(t)})$ increases, and so $L(\theta)$ increases by Jensen's inequality. Obviously, for $s \in \mathcal{S}_L$, $L(\theta)$ increases. Thus $L(\theta)$ increases at each CM-step, and so $L(\theta^{(t+1)}) \geq L(\theta^{(t)})$. $\square$

When the ECME sequence of likelihood values $\{L(\theta^{(t)})\}$ is bounded above, $L(\theta^{(t)})$ converges monotonically to a finite limit L^* . As with EM and ECM, the limit L^* is not necessarily a stationary value of $L(\theta)$. If $\mathcal{H}$ is space-filling at each iteration, we have Theorem 2 that follows from Wu (1983) and Meng & Rubin (1993).

THEOREM 2. Suppose

- (i) $\Theta_{\theta_0} \equiv \{\theta \in \Theta : L(\theta) \geq L(\theta_0)\}$ is compact for any $L(\theta_0) > -\infty$;
- (ii) $L(\theta)$ is continuous in Θ and differentiable in the interior of Θ ;

- (iii) $Q(\theta|\theta')$ and $H(\theta|\theta')$ are continuous in $\Theta \times \Theta$ and differentiable in both θ and θ' in the interior of $\Theta \times \Theta$, and the order of the differentiation and expectation operators involved in $Q(\theta|\theta')$ and $H(\theta|\theta')$ are interchangeable;
- (iv) Θ_{θ_0} is in the interior of Θ for any $\theta_0 \in \Theta$;
- (v) $\mathcal{H} = \{h_s(\theta) : s = 1, \dots, S\}$ is space-filling, where each $h_s(\theta)$ is differentiable, and $\{\nabla h_1(\theta), \dots, \nabla h_S(\theta)\}$ is of full rank at any θ in the interior of Θ ;
- (vi) all S conditional maximisations are unique.

Then all limit points of any instance $\{\theta^{(t)}\}$ of an ECME algorithm are stationary points of $L(\theta)$, and $L(\theta^{(t)})$ converges monotonically to $L^* = L(\theta^*)$ for some stationary point θ^* .

Proof. Following the proof of Theorem 2 and Theorem 3 of Meng & Rubin (1993), the proof of Theorem 2 here is easy because we have

$$D^{10}Q(\theta^{(t)}|\theta^{(t)}) = \partial L(\theta^{(t)})/\partial \theta,$$

which follows from $D^{10}H(\theta^{(t)}|\theta^{(t)}) = 0$ (Dempster et al., 1977). The argument from (4.4) of Meng & Rubin (1993) is still valid with $D^{10}Q(\theta^{(t)}|\theta^{(t)})$ replaced by $\partial L(\theta^{(t)})/\partial \theta$, and thus Theorem 2 follows. $\square$

The convergence of the ECME sequence of likelihood values $\{L(\theta^{(t)})\}$ in general does not imply the convergence of the corresponding ECME sequence $\{\theta^{(t)}\}$. Following Wu (1983, Theorems 4, 5, 6), however, we have results for ECME analogous to those for EM.

As with EM and ECM, the derivative of the $\theta^{(t)} \rightarrow \theta^{(t+1)}$ mapping for ECME at θ^* governs the convergence rate of ECME and can be obtained in terms of the missing-data, observed-data, and complete-data information matrices. The mathematical expressions are tedious, but confirm that typically ECME will converge faster than ECM or EM. The intuition is direct because CM-steps that maximise L rather than Q maximise the correct function rather than a current approximation to it. Of course, a special case of ECME is that all steps maximise $L(\theta)$ with no E-steps, which has quadratic convergence. More precisely, the method of Jamshidian & Jennrich (1993) can be viewed technically as a special case of ECME where each CM-step maximises the actual likelihood and the constraint functions correspond to different conjugate linear combinations of the parameters across iterations.

3. EXAMPLES

In the first example, which involves the multivariate t distribution, the M-step of EM requires iteration, as does the second of two CM-steps of the equivalent ECM algorithm. Maximising the actual likelihood in the second CM-step also requires iteration, but leads to an ECME algorithm that converges substantially faster in both number of iterations and total computer time. The second example concerns factor analysis, where the M-step of EM is in closed form but has been noted for years to lead to rates of convergence so slow that it can be difficult to judge convergence or to have confidence in it due to finite precision for $\theta^{(t)}$. An ECME algorithm, although somewhat more tedious to code since it requires iteration at one of two CM-steps, has a substantially faster rate of convergence, thereby allowing relatively easy and confident assessment of convergence. The third example involves two ECME algorithms for a variance components model. First, we show that the Laird & Ware (1982) 'EM' algorithm is actually an ECME algorithm with closed-form CM-steps. Then, we compare this ECME algorithm with EM and an alternative ECME algorithm with an iterative CM-step that appears to converge more quickly than either EM or the Laird–Ware ECME.

These examples illustrate, first, that in principle ECME can be applied to any problem to which EM or ECM can be applied, secondly, that the rate of convergence of ECME can be substantially faster than either EM or ECM, and thirdly that it is typically easier to judge the convergence of ECME.

Example 3-1: Missing data in the multivariate t . Consider the multivariate t model with centre μ , inner product Ψ , and degrees of freedom ν , $\theta = (\mu, \Psi, \nu)$. For this case, the complete-data are $X = \{X_1, \dots, X_n\}$, where $X_i = (Y_i, \tau_i)$, can be represented as

$$Y_i | (\tau, \theta) \sim N_p(\mu, \Psi/\tau_i) \quad (i = 1, \dots, n)$$

independently, and

$$\tau_i | \theta \sim \chi^2_{\nu}/\nu \quad (i = 1, \dots, n)$$

independently and identically, where the scale factors $\tau = \{\tau_1, \dots, \tau_n\}$ are treated as missing data. When the data matrix Y has missing values Y_{mis} , the partition of X into observed and missing data has Y_{obs} observed and (Y_{mis}, τ) missing.

The EM algorithm for the case where ν is known with no missing data in Y is given by Rubin (1983), and is extended to the case with missing data in Y by Little (1988) and Little & Rubin (1987, pp. 209–16). An EM algorithm for simultaneous maximum likelihood estimation of $\theta = (\mu, \Psi, \nu)$ with missing data in Y is given by Lange, Little & Taylor (1989). Because the expected complete-data loglikelihood is the sum of one term involving only (μ, Ψ) and a second term involving only ν , EM is the same as an ECM algorithm that partitions θ into (μ, Ψ) and ν . Maximising Q over (μ, Ψ) is computationally easy, but the maximisation of Q over ν is not, and Lange et al. (1989) use a one-dimensional search that results in time-consuming M-steps. With fixed μ and Ψ , however, maximising L over ν is still a one-dimensional maximisation, which suggests using ECME because the updating of ν should then be faster, there being no memory of the current estimates of weights, τ , to hinder the appropriate estimation of ν . Only the outlines of the EM = ECM and ECME algorithms are given here; the details of these algorithms for the t family are given by Liu & Rubin (1995).

At iteration $t + 1$:

E-step. Calculate the expected complete-data loglikelihood given current estimates of the parameters, $\mu^{(t)}$, $\Psi^{(t)}$ and $\nu^{(t)}$. The calculations are the same as for EM.

CM-step 1 for ECM and ECME. Fix $\nu = \nu^{(t)}$, and calculate $\mu^{(t+1)}$ and $\Psi^{(t+1)}$ to maximise the expected complete-data likelihood; $\mu^{(t+1)}$ and $\Psi^{(t+1)}$ are in closed form.

CM-step 2 for ECM. Fix $\mu = \mu^{(t+1)}$ and $\Psi = \Psi^{(t+1)}$, and calculate $\nu^{(t+1)}$ to maximise the constrained expected complete-data likelihood, Q , which has a closed form but requires a one-dimensional search to find $\nu^{(t+1)}$.

CM-step 2 for ECME. Fix $\mu = \mu^{(t+1)}$ and $\Psi = \Psi^{(t+1)}$, and calculate $\nu^{(t+1)}$ to maximise the constrained actual likelihood, L , which has a closed form but requires a one-dimensional search to find $\nu^{(t+1)}$.

A multicycle version of ECM (Meng & Rubin, 1993) is obtained by performing a second E-step before the second CM-step to find the expected complete-data loglikelihood given $\theta = (\mu^{(t+1)}, \Psi^{(t+1)}, \nu^{(t)})$. Since the second CM-step of ECME is for the actual likelihood, multicycle ECM in this case is the same as ECME.

As a numerical example, we use the data in Table 1 of Cohen, Dalal & Tukey (1993), which consist of 79 observations of bivariate (x, y) , where we delete the y -value of every third observation starting from $i = 1$, that is $i = 1, 4, 7, \dots$, and the x -value of every third observation starting from $i = 2$, that is $i = 2, 5, 8, \dots$, and treat the corresponding missing-data mechanism as ignorable (Rubin, 1976). We applied $EM = ECM$ and $ECME$ to the resultant data with the bivariate t model and the same starting values of (μ, Ψ) , their maximum likelihood estimate under the bivariate normal, and $\nu = 1000$. The corresponding loglikelihood and ν values at each iteration are displayed in Fig. 1(a) and (b), where the solid line and dashed line represent the values from EM and $ECME$, respectively. The dramatically faster convergence of $ECME$ over EM is obvious, and is so dramatic that the

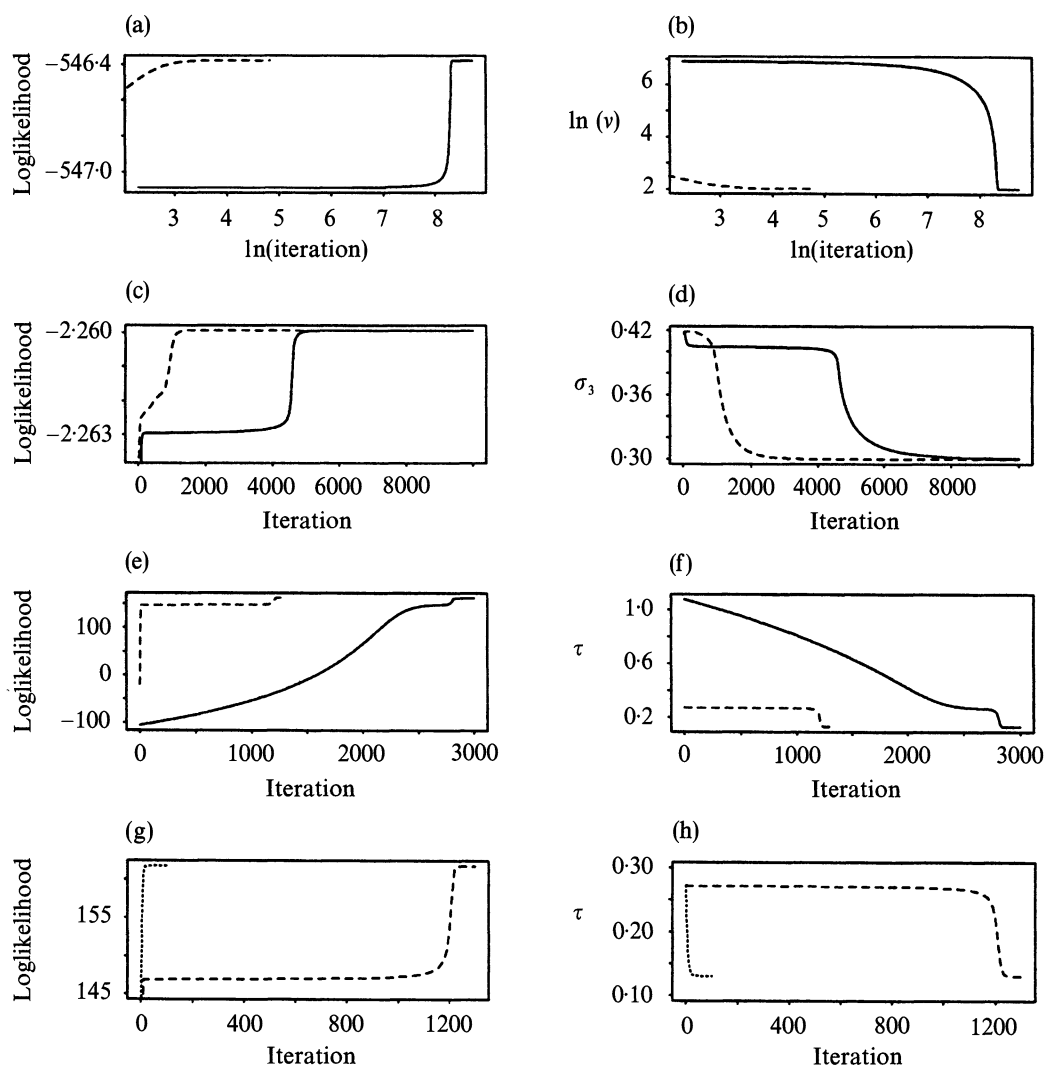


Fig. 1. Convergence of EM (solid line) and $ECME$ (dashed line) in four cases: (a) and (b), Example 3-1, multivariate t ; (c) and (d), Example 3-2, factor analysis; (e) and (f), Example 3-3, variance components; (g) and (h) are for Example 3-3 but compare the version of $ECME$ from (e) and (f) (dashed line) with another version of $ECME$ (dotted line). (a), (c), (e) and (g) display convergence of likelihoods; (b), (d), (f) and (h) display selected parameters.

fact that the starting values are the same is lost in the plots, which display results as functions of log number of iterations. The total CPU time until convergence for ECME is nearly one-tenth that for EM in this example (Lui & Rubin, 1995, Table 3).

Example 3.2. Factor analysis. Factor analysis can be viewed as the normal linear regression analysis of an observable p -dimensional variable Y on an unobservable variable Z consisting of $q < p$ factors, which are themselves normal; the key assumption is that the components of Y are conditionally independent given Z . Commonly, the factors are assumed orthonormal. For n independent observations of Y , we then have, first, given $Z_1, \dots, Z_n$ and (β, σ^2) , $Y_1, \dots, Y_n$ are independent with

$$Y_i | (Z_i, \beta, R, \sigma^2) \sim N_p(Z_i \beta, \text{Diag}(\sigma_1^2, \dots, \sigma_p^2)) \quad (i = 1, \dots, n),$$

where β is called the factor-loading matrix, $\sigma^2 = (\sigma_1^2, \dots, \sigma_p^2)$ is called the vector of uniquenesses, and secondly, given (β, σ^2) , $Z_1, \dots, Z_n$ are independently and identically distributed with $Z_i \sim N_q(0, I)$ ($i = 1, \dots, n$). The complete data are $X = (Y, Z)$, and Z is fully missing.

Dempster et al. (1977) state that the EM algorithm can be used to find maximum likelihood estimates of the parameters, and Rubin & Thayer (1982) provide the detailed algorithm. Briefly, it is as follows where the complete-data sufficient statistics for (β, σ^2) are

$$C_{yy} = \sum_{i=1}^n (Y_i - \bar{Y})(Y_i - \bar{Y})/n, \quad C_{yz} = \sum_{i=1}^n (Y_i - \bar{Y})'Z_i/n, \quad C_{zz} = \sum_{i=1}^n Z_i'Z_i/n.$$

E-step for EM. Given the current estimate of $\theta = (\beta, \sigma^2)$, calculate the expected complete-data sufficient statistics

$$E(C_{yy} | Y, \theta) = C_{yy}, \quad E(C_{yz} | Y, \theta) = C_{yy}\gamma, \quad E(C_{zz} | Y, \theta) = \gamma' C_{yy} \gamma + \Delta,$$

where γ and Δ are the regression coefficients and residual covariance matrix of Z on Y , which are functions of $\theta = (\beta, \sigma^2)$.

M-step for EM. Replace the complete-data sufficient statistics, C_{yy} , C_{yz} and C_{zz} , with their expected values and find the complete-data maximum likelihood estimate of $\theta = (\beta, \sigma^2)$. For example, with no constraint on β ,

$$\hat{\beta} = (\gamma' C_{yy} \gamma + \Delta)^{-1} \gamma' C_{yy}, \quad \hat{\sigma}^2 = \text{diag} \{C_{yy} - C_{yy} \gamma (\gamma' C_{yy} \gamma + \Delta)^{-1} \gamma' C_{yy}\}.$$

Because of the large fraction of missing information in the missing factor scores Z , EM can converge very slowly. Thus we consider ECME, where we partition $\theta = (\beta, \sigma^2)$ into $\theta_1 = \beta$ and $\theta_2 = \sigma^2$ with $\mathcal{S}_Q = \{1\}$ and $\mathcal{S}_L = \{2\}$. The reason for this choice is that the actual likelihood is simply a p -dimensional normal with restrictions on the covariance matrix, and numerically maximising $L(\theta)$ over p -dimensional σ^2 is substantially easier than numerically maximising $L(\theta)$ over $p \times q$ -dimensional β , or maximising $L(\theta)$ simultaneously over σ^2 and β . Thus, each iteration of this version consists of an E-step and two CM-steps.

E-step for ECME. This is the same as the E-step of EM.

CM-step 1 for ECME. This is the same as the M-step of EM for finding $\hat{\beta}$.

CM-step 2 for ECME. Find $\hat{\sigma}^2$ to maximise the actual constrained likelihood given β , which can be done, for example, using Newton–Raphson iterations.

We applied both EM and ECME to the following data and model from Rubin & Thayer (1982) with $p = 9$:

$$C_{yy} = \begin{pmatrix} 1.0 & 0.554 & 0.227 & 0.189 & 0.461 & 0.506 & 0.408 & 0.280 & 0.241 \\ & 1.0 & 0.296 & 0.219 & 0.479 & 0.530 & 0.425 & 0.311 & 0.311 \\ & & 1.0 & 0.769 & 0.237 & 0.243 & 0.304 & 0.718 & 0.730 \\ & & & 1.0 & 0.212 & 0.226 & 0.291 & 0.681 & 0.661 \\ & & & & 1.0 & 0.520 & 0.514 & 0.313 & 0.245 \\ & & & & & 1.0 & 0.473 & 0.348 & 0.290 \\ & & & & & & 1.0 & 0.374 & 0.306 \\ & & & & & & & 1.0 & 0.672 \\ & & & & & & & & 1.0 \end{pmatrix}.$$

In the model, $q = 4$ and, a priori, there are zero factor loadings on factor 4 for variables 1–4 and zero factor loadings on factor 3 for variables 5–9. With the same starting values of β created from the principal components and eigenvalues of C_{yy} and the same small starting values of σ^2 , ECME appears to converge much more rapidly than EM, as shown in Fig. 1(c) and (d), where the corresponding loglikelihood functions and the estimates of σ_3^2 are displayed. In terms of computer time, ECME appears to be only 25% faster than EM in this example, although the assessment of convergence is substantially easier with ECME.

Example 3.3: A variance components model. As our final example, we consider the following variance components, or repeated measures, model (Hartley & Rao, 1967; Harville, 1977; Laird & Ware, 1982; Laird et al., 1987):

$$y_i = X_i\beta + Z_ib_i + e_i, \quad (2)$$

for $i = 1, \dots, m$, where y_i is $(n_i \times 1)$ observed vector, X_i and Z_i are known $(n_i \times p)$ and $(n_i \times q)$ design matrices, respectively, $n = \sum_{i=1}^m n_i$, β is a $(p \times 1)$ fixed effect, b_i is the $(q \times 1)$ random effect, $b_i \sim N_q(0, \Psi)$, b_i and e_i are independent, and $e_i \sim N_{n_i}(0, R_i\sigma^2)$ with known $R_i > 0$, unknown $(q \times q)$ positive definite matrix Ψ , and unknown positive scalar σ^2 . Also, (y_i, b_i, e_i) are independent for $i = 1, \dots, m$. To find the maximum likelihood estimate of $\theta = (\beta, \Psi, \sigma^2)$, following Dempster et al. (1977), Laird & Ware (1982) and Laird et al. (1987), we treat $b = (b_1, \dots, b_m)$ as missing data. The complete data thus consist of $Y = (y_1, \dots, y_m)$ and b .

Routine algebraic operations lead to the following standard EM algorithm.

E-step. Using current estimates of $\theta = (\beta, \Psi, \sigma^2)$, calculate, for $i = 1, \dots, m$,

$$\hat{b}_i = E(b_i | Y, \beta, \Psi, \sigma^2) = (\sigma^2\Psi^{-1} + Z_i'R_i^{-1}Z_i)^{-1}Z_i'R_i^{-1}(y_i - X_i\beta),$$

$$\hat{\Psi}_i = E(b_ib_i' | Y, \beta, \Psi, \sigma^2) = \hat{b}_i\hat{b}_i' + (\Psi^{-1} + Z_i'R_i^{-1}Z_i\sigma^{-2})^{-1}.$$

M-step. Calculate

$$\begin{aligned}\hat{\beta} &= \left(\sum_{i=1}^m X_i' R_i^{-1} X_i \right)^{-1} \sum_{i=1}^m X_i' R_i^{-1} (y_i - Z_i \hat{b}_i), \quad \hat{\Psi} = \frac{1}{m} \sum_{i=1}^m \hat{\Psi}_i, \\ \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^m [(y_i - X_i \hat{\beta} - Z_i \hat{b}_i)' R_i^{-1} (y_i - X_i \hat{\beta} - Z_i \hat{b}_i) \\ &\quad + \text{tr} \{ Z_i' R_i^{-1} Z_i (\Psi^{-1} + Z_i' R_i^{-1} Z_i \sigma^{-2})^{-1} \}],\end{aligned}\quad (3)$$

which maximises the expected complete-data loglikelihood.

There are different versions of ECME for model (2). We consider two versions that can be especially appropriate when q , the dimension of Ψ is relatively large. First, partition θ into $\theta_1 = (\Psi, \sigma^2)$ and $\theta_2 = \beta$ with $\mathcal{S}_Q = \{1\}$ and $\mathcal{S}_L = \{2\}$.

ECME version 1

E-step. This is the same as the E-step of EM.

CM-step 1. Calculate $\hat{\Psi}$ and $\hat{\sigma}^2$ as in the M-step of EM.

CM-step 2. Calculate

$$\hat{\beta} = \left\{ \sum_{i=1}^m X_i' (Z_i \hat{\Psi} Z_i' + R_i \hat{\sigma}^2)^{-1} X_i \right\}^{-1} \left\{ \sum_{i=1}^m X_i' (Z_i \hat{\Psi} Z_i' + R_i \hat{\sigma}^2)^{-1} y_i \right\},$$

which maximises the constrained actual likelihood over β given $\Psi = \hat{\Psi}$ and $\sigma^2 = \hat{\sigma}^2$.

This version of ECME, in fact, is the algorithm given by Laird & Ware (1982) and Laird et al. (1987), who mistakenly call it EM. The algorithm was called a hybrid EM algorithm by Jennrich & Schluchter (1986), who also realised it is not EM.

In version 2 of ECME for this model, partition θ into $\theta_1 = \Psi$, $\theta_2 = \beta$ and $\theta_3 = \sigma^2$ with $\mathcal{S}_Q = \{1\}$ and $\mathcal{S}_L = \{2, 3\}$. Notice that, in contrast to version 1, the maximisation of L is over β and also σ^2 , and although the latter maximisation requires iteration, it is only a one-dimensional maximisation.

ECME version 2

E-step. This is the same as the E-step of EM.

CM-step 1. Calculate $\hat{\Psi}$ as in the M-step of EM.

CM-step 2. Calculate $\hat{\beta}$ as in CM-step 2 of ECME version 1.

CM-step 3. Calculate $\hat{\sigma}^2$ to maximise the constrained actual loglikelihood

$$-\frac{1}{2} \sum_{i=1}^m \log |R_i \sigma^2 + Z_i \Psi Z_i'| - \frac{1}{2} \sum_{i=1}^m (y_i - X_i \hat{\beta})' (R_i \sigma^2 + Z_i \Psi Z_i')^{-1} (y_i - X_i \hat{\beta})$$

given $\beta = \hat{\beta}$ and $\Psi = \hat{\Psi}$, which can be done using Newton–Raphson iterations.

As a numerical example, we use the data of Besag (1991), which consist of the layout and yields for 62 varieties of winter wheat in three physically separated complete replicates, and use the first-difference Gaussian noise model for the random fertility effects. That is, we have $m = 3$ and $n_i = 61$, where y_i gives the 61 differences of the yields in replicate i , X_i is the corresponding (61×62) design matrix, $Z_i = I_{61 \times 61}$, $R_i = R$ is the (61×61) Toeplitz matrix with the first row $(2, -1, 0, \dots, 0)$, and β is the fixed variety effects with the linear constraint $\sum \beta_j = 0$. Instead of an unrestricted covariance matrix Ψ , here we have $\Psi = I\tau^2$, where $\tau^2 \geq 0$ is unknown scalar, so that the parameters are now simply $\theta = (\beta, \tau^2, \sigma^2)$.

With (3) replaced by

$$\hat{\tau}^2 = \frac{1}{n} E \left(\sum_{i=1}^m b_i' b_i \middle| \beta, \Psi, \sigma^2 \right) = \frac{1}{n} \sum_{i=1}^m [\hat{b}_i' \hat{b}_i + \text{tr} \{ (R\sigma^2 + I\tau^2)^{-1} R\sigma^2 \tau^2 \}], \quad (4)$$

the E-step and M-step(s) of EM and ECME remain the same as those for the model with an unrestricted $(q \times q)$ covariance matrix Ψ .

We applied EM, ECME version 1, and ECME version 2 with the following common starting value: $\beta = 0$, $\tau^2 = (1/n) \sum_{i=1}^m y_i y_i$ and $\sigma^2 = \tau^2 \times 10^{-4}$. The corresponding loglikelihood and τ values are displayed in Fig. 1, where, in Fig. 1(e) and (f), the solid line corresponds to EM and the dashed line corresponds to ECME version 1, and, in Fig. 1(g) and (h), the dashed line corresponds to ECME version 1 and the dotted line corresponds to ECME version 2, using an expanded scale. Clearly, at least in terms of number of iterations, ECME version 2 is substantially faster than ECME version 1, which itself is substantially faster than EM. In terms of total computer time in this example, ECME-2 is approximately twice as fast as EM, which itself is approximately an order of magnitude faster than ECME-1.

4. THEORY ON RATES OF CONVERGENCE

Suppose that the sequence $\{\theta^{(t)}\}$ from an iterative algorithm is generated by the mapping $M(\theta)$, and that $\theta^{(t)} \rightarrow \theta^*$ as $t \rightarrow \infty$. Following Meng (1994), we refer to $DM(\theta^*)$, the derivative of $M(\theta)$ at $\theta = \theta^*$, as the matrix rate, or simply the rate, of convergence of the sequence $\theta^{(t)}$ for the corresponding algorithm. Also, $I - DM(\theta^*)$ is referred to as the matrix speed, or simply the speed, of convergence of $\theta^{(t)}$. The maximum absolute eigenvalue of $DM(\theta^*)$, $\lambda_{\max}\{DM(\theta^*)\}$, is often known as the global rate of convergence of $\theta^{(t)}$. Similarly, $1 - \lambda_{\max}\{DM(\theta^*)\}$ is known as the global speed of convergence of the sequence $\theta^{(t)}$ (Meng, 1994).

We denote the EM, ECM, ECME, and CM, i.e. iterative conditional modes (Besag, 1986), mappings as $\theta^{(t+1)} = M^{\text{EM}}(\theta^{(t)})$, $\theta^{(t+1)} = M^{\text{ECM}}(\theta^{(t)})$, $\theta^{(t+1)} = M^{\text{ECME}}(\theta^{(t)})$ and $\theta^{(t+1)} = M^{\text{CM}}(\theta^{(t)})$, respectively. The rates and speeds of convergence of EM, ECM and ECME are closely related to the missing-data, complete-data and observed-data information matrices:

$$I_{\text{mis}} = -D^{20} H(\theta^* | \theta^*), \quad (5)$$

$$I_{\text{com}} = -D^{20} Q(\theta^* | \theta^*), \quad (6)$$

$$I_{\text{obs}} = I_{\text{com}} - I_{\text{mis}} = -D^2 L(\theta^*). \quad (7)$$

Dempster et al. (1977) show, for example, that under mild conditions the rate of convergence of the EM algorithm is

$$DM^{\text{EM}}(\theta^*) = I_{\text{mis}} I_{\text{com}}^{-1} = I_{\text{mis}} (I_{\text{mis}} + I_{\text{obs}})^{-1}, \quad (8)$$

which provides ‘the fractions of missing information’. The speed of convergence of EM is thus

$$\text{speed of EM} = I_{\text{obs}} I_{\text{com}}^{-1} = I_{\text{obs}} (I_{\text{mis}} + I_{\text{obs}})^{-1},$$

which provides ‘the fractions of observed information’. For the ECM algorithm, Meng (1994) shows that, under mild conditions,

$$\text{speed of ECM} = \text{speed of EM} \times \text{speed of CM with complete data}.$$

The rate of convergence of the ECME algorithm can be obtained similarly. The algebraic

expression is tedious. Here, we present only the rate of convergence of ECME for the special but common case when the parameter constraints are formed by creating a partition, $\mathcal{P}$, of θ as $\theta = (\theta_1, \dots, \theta_S)$ with associated dimensions $d_1, \dots, d_S$; that is, for $s = 1, \dots, S$, $h_s(\theta) = (\theta_1, \dots, \theta_{s-1}, \theta_{s+1}, \dots, \theta_S)$. In this case, (1), with θ' replaced by ϕ , can be written as

$$L(\theta_1, \dots, \theta_S) = Q(\theta_1, \dots, \theta_S | \phi_1, \dots, \phi_S) - H(\theta_1, \dots, \theta_S | \phi_1, \dots, \phi_S). \quad (9)$$

Let $D^{(i)} L(\theta_1, \dots, \theta_S)$ denote $\partial L(\theta) / \partial \theta_i$ and $D^{(i,j)} L(\theta_1, \dots, \theta_S)$ denote $\partial^2 L(\theta) / (\partial \theta_i \partial \theta_j)$. Similar notation applies to both $Q(\theta_1, \dots, \theta_S | \phi_1, \dots, \phi_S)$ and $H(\theta_1, \dots, \theta_S | \phi_1, \dots, \phi_S)$ where the number of vector variables is extended to $2S$. However, for simplicity, D^{10} , D^{20} and D^{11} follow the notation of Dempster et al. (1977).

For the partition $\mathcal{P}$, the version of ECME with $\mathcal{S}_Q = \{1, \dots, S_0\}$ and $\mathcal{S}_L = \{S_0 + 1, \dots, S\}$ has a relatively neat expression for $DM^{\text{ECME}}(\theta^*)$ derived in Theorem 3. Different forms of $DM^{\text{ECME}}(\theta^*)$ are obtained for the different versions of ECME that are defined by different orderings of the S steps, and typically the corresponding notation becomes more tedious.

THEOREM 3. Suppose

- (i) $Q(\theta | \phi)$, $L(\theta)$ and $H(\theta | \phi)$ are twice differentiable in the interior of Θ and the order of the related differentiation and expectation operators are interchangeable;
- (ii) $\theta^{(t)} \rightarrow \theta^* \in$ the interior of Θ ;
- (iii) $D^{\{s\}} Q(\theta_1^{(t+1)}, \dots, \theta_s^{(t+1)}, \theta_{s+1}^{(t)}, \dots, \theta_S^{(t)} | \theta_1^{(t)}, \dots, \theta_S^{(t)}) = 0 \quad (1 \leq s \leq S_0),$
 $D^{\{s\}} L(\theta_1^{(t+1)}, \dots, \theta_s^{(t+1)}, \theta_{s+1}^{(t)}, \dots, \theta_S^{(t)}) = 0 \quad (S_0 < s \leq S).$

Then

$$DL(\theta^*) = 0, \quad (10)$$

$$DM^{\text{ECME}}(\theta^*) = I - I_{\text{obs}}(T_{I_{\text{com}}, \mathcal{P}} - T_{I_{\text{mis}}, \mathcal{P}} J_{\mathcal{P}})^{-1}, \quad (11)$$

where $T_{A, \mathcal{P}}$ denotes the block upper triangular part of a symmetric matrix A with respect to the partition $\mathcal{P}$, $\theta = (\theta_1, \dots, \theta_S)$, and

$$J_{\mathcal{P}} = \begin{pmatrix} 0 & 0 \\ 0 & I_{d_{S_0+1} + \dots + d_S} \end{pmatrix}.$$

The proof of Theorem 3 is given in the Appendix.

Let $S_0 = S$, that is $\mathcal{S}_L = \emptyset$; then (11) becomes

$$DM^{\text{ECM}}(\theta^*) = I - I_{\text{obs}} T_{I_{\text{com}}, \mathcal{P}}^{-1}, \quad (12)$$

which is obtained by Meng in his Ph.D. thesis. Letting $S_0 = S$ and $d_1 = d$, (12) becomes (8), that is $DM^{\text{EM}}(\theta^*) = I_{\text{mis}} I_{\text{com}}^{-1}$.

For $S_0 = 0$, from (7) and (11), we have

$$DM^{\text{CM}}(\theta^*) = I - I_{\text{obs}} T_{I_{\text{obs}}, \mathcal{P}}^{-1}. \quad (13)$$

When $X = Y$, that is there are no missing data, (13) is equivalent to

$$DM^{\text{CM}}(\theta^*) = I - I_{\text{com}} T_{I_{\text{com}}, \mathcal{P}}^{-1}, \quad (14)$$

which is derived in Meng's Ph.D. thesis from (12).

In term of speed of convergence, we have

$$\begin{aligned} \text{speed of ECM} &= I_{\text{obs}} T_{I_{\text{com}}, \mathcal{P}}^{-1}, \\ \text{speed of CM} &= \begin{cases} I_{\text{obs}} T_{I_{\text{obs}}, \mathcal{P}}^{-1} & \text{with missing data,} \\ I_{\text{com}} T_{I_{\text{com}}, \mathcal{P}}^{-1} & \text{with complete data, that is } X = Y. \end{cases} \end{aligned} \quad (15)$$

Therefore, the speeds of convergence of EM, ECM and CM have the following relationship:

$$\text{speed of ECM} = \text{speed of EM} \times \text{speed of CM with complete data}.$$

When $\mathcal{S}_Q = \emptyset$, we obtain

$$\text{speed of ECM} = \text{speed of CM with incomplete data} \times (T_{I_{\text{com}}, \mathcal{S}} - T_{I_{\text{mis}}, \mathcal{S}}) T_{I_{\text{com}}, \mathcal{S}}^{-1}.$$

We call $(T_{I_{\text{com}}, \mathcal{S}} - T_{I_{\text{mis}}, \mathcal{S}}) T_{I_{\text{com}}, \mathcal{S}}^{-1}$ ‘the sequential fractions of observed information’ since it involves only the block upper triangular parts, which correspond to the ordering of the S maximisations, of I_{com} and I_{mis} . Accordingly, we call $(T_{I_{\text{com}}, \mathcal{S}} - T_{I_{\text{mis}}, \mathcal{S}} J_{\mathcal{S}}) T_{I_{\text{com}}, \mathcal{S}}^{-1}$ ‘the partial fractions of observed information’. From (11) and (15), we have

$$\text{speed of ECM} = \text{speed of ECME} \times (T_{I_{\text{com}}, \mathcal{S}} - T_{I_{\text{mis}}, \mathcal{S}} J_{\mathcal{S}}) T_{I_{\text{com}}, \mathcal{S}}^{-1};$$

that is

$$\text{speed of ECM} = \text{speed of ECME} \times \text{the partial fractions of observed information}.$$

Once again, we see that ECME typically has a greater speed of convergence than ECM. However, there exist cases where the global speed of convergence of ECME is slower than that of ECM.

Table 1. *Matrices for calculating the theoretical large-sample rates of convergence of EM and ECME for the univariate t distribution model with centre μ , scale σ , and degrees of freedom ν to be estimated*

Matrix	Expression
$\lim_{n \rightarrow \infty} \frac{1}{n} I_{\text{com}}$	$\text{diag} \left\{ \sigma^{-2}, \frac{\sigma^4}{2}, \psi' \left(\frac{\nu}{2} \right) - \frac{2}{\nu} \right\}$
$\lim_{n \rightarrow \infty} \frac{1}{n} I_{\text{obs}}$	$\begin{pmatrix} \frac{\nu+1}{\nu+3} \sigma^{-2} & 0 & 0 \\ 0 & \frac{\nu \sigma^4}{2(\nu+3)} & \frac{2\sigma^2}{(\nu+1)(\nu+3)} \\ 0 & \frac{2\sigma^2}{(\nu+1)(\nu+3)} & \psi' \left(\frac{\nu}{2} \right) - \psi' \left(\frac{\nu+1}{2} \right) - \frac{2(\nu+5)}{\nu(\nu+1)(\nu+3)} \end{pmatrix}$
$\lim_{n \rightarrow \infty} DM^{\text{EM}}(\theta^*)$	$\begin{pmatrix} \frac{2}{\nu+3} & 0 & 0 \\ 0 & \frac{3}{\nu+3} & -\frac{2\sigma^2}{(\nu+1)(\nu+3)} \left\{ \psi' \left(\frac{\nu}{2} \right) - \frac{2}{\nu} \right\}^{-1} \\ 0 & -\frac{4}{\sigma^2(\nu+1)(\nu+3)} \left\{ \psi' \left(\frac{\nu+1}{2} \right) - \frac{2(\nu^2+3\nu-2)}{\nu(\nu+1)(\nu+3)} \right\} \left\{ \psi' \left(\frac{\nu}{2} \right) - \frac{2}{\nu} \right\}^{-1} \end{pmatrix}$
$\lim_{n \rightarrow \infty} DM^{\text{ECME}}(\theta^*)$	$\begin{pmatrix} \frac{2}{\nu+3} & 0 & 0 \\ 0 & \frac{3}{\nu+3} & -\frac{6\sigma^2}{(\nu+1)(\nu+3)^2} \left\{ \psi' \left(\frac{\nu}{2} \right) - \psi' \left(\frac{\nu+1}{2} \right) - \frac{2(\nu+5)}{\nu(\nu+1)(\nu+3)} \right\}^{-1} \\ 0 & -\frac{4}{\sigma^2(\nu+1)(\nu+3)} & \frac{8}{(\nu+1)^2(\nu+3)^2} \left\{ \psi' \left(\frac{\nu}{2} \right) - \psi' \left(\frac{\nu+1}{2} \right) - \frac{2(\nu+5)}{\nu(\nu+1)(\nu+3)} \right\}^{-1} \end{pmatrix}$

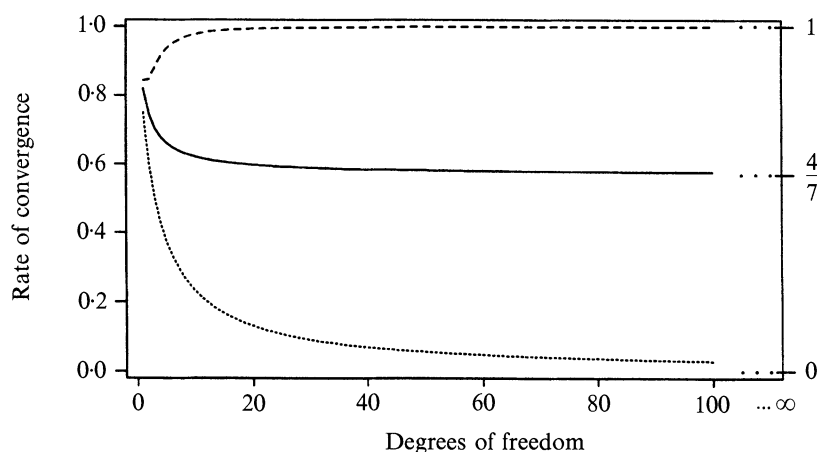


Fig. 2. The large-sample convergence rates of EM and ECME for the univariate $t_1(\mu, \sigma^2, \nu)$ distribution; solid line, ECME with unknown degrees of freedom, ν ; dashed line, EM with unknown degrees of freedom, ν ; dotted line, EM with known degrees of freedom, ν .

To illustrate the results, consider n observations from the univariate t distribution with centre μ , scale σ , and degrees of freedom ν . Differentiating the Q and L functions with respect to the parameters $\theta = (\mu, \sigma^2, \nu/2)$ and letting $\psi(\cdot)$ be the Digamma function, see, for example, Davis (1965), we obtain expressions for

$$\lim_{n \rightarrow \infty} \frac{1}{n} I_{\text{com}}, \quad \lim_{n \rightarrow \infty} \frac{1}{n} I_{\text{obs}},$$

and hence for $\lim_{n \rightarrow \infty} DM^{\text{EM}}(\theta^*)$ and $\lim_{n \rightarrow \infty} DM^{\text{ECME}}(\theta^*)$, as given in Table 1. If ν is known, we can see that the large-sample rate of convergence of EM is $3/(\nu + 3)$, which is also obtained by Dempster, Laird & Rubin (1980). Meng & Rubin (1994) give similar calculations for a contamination model.

The large-sample rates of convergence of EM and ECME with ν to be estimated are the maximum eigenvalues of $\lim_{n \rightarrow \infty} DM^{\text{EM}}(\theta^*)$ and $\lim_{n \rightarrow \infty} DM^{\text{ECME}}(\theta^*)$, which are the solutions to quadratic equations. These solutions are smooth functions of the true ν , as displayed in Fig. 2 along with their large ν limiting values. We have that ECME converges much more quickly than EM for the t distribution with unknown ν , especially when ν is large. The reason that EM theoretically slows down to zero speed when estimating the unknown degrees of freedom with underlying normal data is that the true value of the parameter, $\nu = \infty$, is on the boundary of the parameter space and infinitely far from any starting point in the interior of the parameter space.

5. RELATIONSHIPS WITH MARKOV CHAIN MONTE CARLO METHODS

Markov chain Monte Carlo methods are now widely used for calculating posterior distributions, especially for complicated models; see, for example, Gelfand & Smith (1990), Gelfand et al. (1990), Gelman & Rubin (1992), Besag & Green (1993), Gilks et al. (1993) and Smith & Roberts (1993). As emphasised by Gelman & Rubin (1992), it is wise to find maximum likelihood estimates or posterior modes of the parameters before using iterative simulations because of difficulties assessing convergence of iterative simulation

methods, particularly when the regions of high density are not a priori known. Therefore, EM, ECM and ECME algorithms can be very useful for starting iterative simulations, especially because they are conceptually very similar to Markov chain Monte Carlo methods, simply substituting expectation and maximisation operators for random drawing operators.

Tanner & Wong (1987) clearly described the relationship between the EM algorithm and the data augmentation algorithm: the data augmentation algorithm can be viewed as a stochastic version of EM, where the E-step and the M-step of EM correspond to the imputation step and the posterior step of the data augmentation algorithm, respectively. The analogous relationship between the extension of the data augmentation algorithm, the Gibbs sampler, and the extension of EM, the ECM algorithm, is immediate and made by Meng & Rubin (1992).

As we have seen, because the rates of convergence of EM and ECM are governed by the fractions of missing information, they are typically slower than the ECME algorithm, which allows missing data to be integrated out when maximising a constrained likelihood function. This integration is analogous to different groupings of random variables in the Gibbs sampler, and this analogy can help to clarify the source of the expected gains with ECME. Consider the simple case with missing random variable Y_{mis} and random parameter $\theta = (\theta_1, \theta_2)$ whose mode we seek: ECM employs an E-step using the distribution $(Y_{\text{mis}}|\theta_1, \theta_2)$, a CM-step 1 using the distribution $(\theta_1|Y_{\text{mis}}, \theta_2)$, and a CM-step 2 using the distribution $(\theta_2|Y_{\text{mis}}, \theta_1)$; the analogous Gibbs sampler takes draws from these three conditional distributions. In contrast, ECME uses the same E-step and the same CM-step 1, but uses the distribution $(\theta_2|\theta_1)$ for the second CM-step; the analogous Gibbs sampler takes draws from these latter three distributions, a procedure that is equivalent to the two-step Gibbs sampler drawing $(Y_{\text{mis}}, \theta_2|\theta_1)$ in one step and $(\theta_1|Y_{\text{mis}}, \theta_2)$ in the other. An example of this is given by Liu (1995) for posterior simulation of the multivariate t distribution. Liu, Wong & Kong (1994) and Liu (1994) give results supporting the expected more rapid convergence with the two-step Gibbs sampler.

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APPENDIX

Proof of Theorem 3

For $1 \leq i \leq S_0$, since

$$D^{(i)}L(\theta_1, \dots, \theta_S) = D^{(i)}Q(\theta_1, \dots, \theta_S|\phi_1, \dots, \phi_S) - D^{(i)}H(\theta_1, \dots, \theta_S|\phi_1, \dots, \phi_S), \quad D^{(10)}H(\theta|\theta) = 0$$

(Dempster et al., 1977, eqn. (3.15)), letting $\theta = (\theta_1^{(t+1)}, \dots, \theta_{i+1}^{(t+1)}, \theta_i^{(t)}, \dots, \theta_S^{(t)})$, $\phi = \theta^{(t)}$ and $t \rightarrow \infty$, we have $D^{(i)}L(\theta_1^*, \dots, \theta_S^*) = 0$. For $S_0 < i \leq S$, letting $\theta = (\theta_1^{(t+1)}, \dots, \theta_i^{(t+1)}, \theta_{i+1}^{(t)}, \dots, \theta_S^{(t)})$, we get

$$D^{(i)}L(\theta_1^*, \dots, \theta_S^*) = \lim_{t \rightarrow \infty} D^{(i)}L(\theta_1^{(t+1)}, \dots, \theta_i^{(t+1)}, \theta_{i+1}^{(t)}, \dots, \theta_S^{(t)}) = 0.$$

Therefore (10) is proved.

For $1 \leq i \leq S_0$, expanding $D^{(i)}Q(\theta_1, \dots, \theta_S|\phi_1, \dots, \phi_S)$ at $(\theta, \phi) = (\theta^*, \theta^*)$ and letting $\theta =$

$(\theta_1^{(t+1)}, \dots, \theta_i^{(t+1)}, \theta_{i+1}^{(t)}, \dots, \theta_S^{(t)})$ and $\phi = \theta^{(t)}$, we have

$$0 = (\theta^{(t+1)} - \theta^*) \begin{pmatrix} D^{\{i,1\}} Q(\theta^* | \theta^*) \\ \vdots \\ D^{\{i,i\}} Q(\theta^* | \theta^*) \\ 0 \\ \vdots \\ 0 \end{pmatrix} + (\theta^{(t)} - \theta^*) \begin{pmatrix} D^{\{i,S+1\}} Q(\theta^* | \theta^*) \\ \vdots \\ D^{\{i,S+i\}} Q(\theta^* | \theta^*) \\ D^{\{i,S+i+1\}} Q(\theta^* | \theta^*) + D^{\{i,i+1\}} Q(\theta^* | \theta^*) \\ \vdots \\ D^{\{i,S+S\}} Q(\theta^* | \theta^*) + D^{\{i,S\}} Q(\theta^* | \theta^*) \end{pmatrix} + \dots \quad (\text{A1})$$

For $S_0 + 1 \leq i \leq S$, expanding $D^{\{i\}} L(\theta_1, \dots, \theta_S)$ at $\theta = \theta^*$ and letting

$$\theta = (\theta_1^{(t+1)}, \dots, \theta_i^{(t+1)}, \theta_{i+1}^{(t)}, \dots, \theta_S^{(t)}),$$

we have

$$0 = (\theta^{(t+1)} - \theta^*) \begin{pmatrix} D^{\{i,1\}} L(\theta^*) \\ \vdots \\ D^{\{i,i\}} L(\theta^*) \\ 0 \\ \vdots \\ 0 \end{pmatrix} + (\theta^{(t)} - \theta^*) \begin{pmatrix} 0 \\ \vdots \\ 0 \\ D^{\{i,i+1\}} L(\theta^*) \\ \vdots \\ D^{\{i,S\}} L(\theta^*) \end{pmatrix} + \dots \quad (\text{A2})$$

Since $M^{\text{ECME}}(\theta^*) = \theta^*$ and $M^{\text{ECME}}(\theta^{(t)}) = \theta^{(t+1)}$,

$$\theta^{(t+1)} - \theta^* = M^{\text{ECME}}(\theta^{(t)}) - M^{\text{ECME}}(\theta^*) = (\theta^{(t)} - \theta^*) DM^{\text{ECME}}(\theta^*) + \dots \quad (\text{A3})$$

Equation (1) and formula (3.16) of Dempster et al. (1977) imply

$$D^{11} Q(\theta^* | \theta^*) = D^{11} H(\theta^* | \theta^*) = -D^{20} H(\theta^* | \theta^*), \quad (\text{A4})$$

$$D^2 L(\theta^*) = D^{20} Q(\theta^* | \theta^*) - D^{20} H(\theta^* | \theta^*). \quad (\text{A5})$$

Combining (A1) and (A2) and using (A3), (A4), (A5) and notations I_{com} and I_{mis} in (6) and (5), respectively, in the limit of $t \rightarrow \infty$ equation (11) follows. $\square$

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